14POINT7
CUTTING EDGE IN MOTION

SPARTAN 3 V2

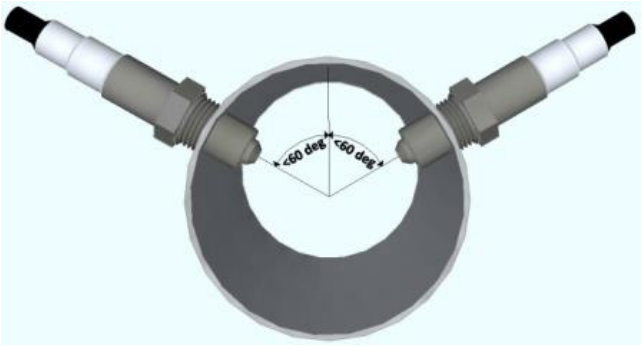
1. Warning

- **Do not connect or disconnect the Lambda Sensor while Spartan 3 is powered.**
- **The Lambda Sensor will get very hot during normal operation, please be careful when handling it.**
- **Do not install the Lambda Sensor in such a manner that the unit is powered before your engine is running. An engine start can move condensation in your exhaust system to the sensor, if the sensor is already heated this can cause thermal shock and cause the ceramic internals inside the sensor to crack and deform.**
- **While the Lambda Sensor is in an active exhaust stream, it must be controlled by Spartan 3. Carbon from an active exhaust can easily build up on an unpowered sensor and foul it.**
- **Lambda sensor life when used with leaded fuels is between 100-500 hrs.**

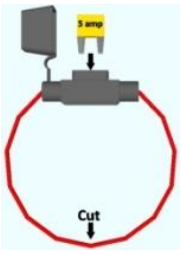
2. Package Contents

1x Spartan 3 v2, 1x USB Converter, 1x Waterproof blade fuse holder, 2x 5 Amp blade fuse

3. Exhaust Installation



The Lambda sensor should be installed between the 10 o'clock and the 2 o'clock position, less than 60 degrees from vertical, this will allow gravity to remove water condensation from the sensor.
For all Oxygen sensor installations, the sensor must be installed before the catalytic converter.
For normally aspirated engines the sensor should be installed about 2ft from the engine exhaust port. For Turbocharged engines the sensor should be installed after the turbocharger. For Supercharged engines the sensor should be installed 3ft from the engine exhaust port.



4. Fuse

Insert 5 amp fuse into fuse holder, cut wire at midpoint, and secure lid. One end of the fuse holder connects to Spartan 3 red wire the other end of the fuse holder connects to a switched 12[v] source, fuel pump relay is normally recommended.

5.Wiring

Wire Color	Name	Connects to	Note
Red	Power	Switched 12[v]	Use fuse holder, 12[v] should be live only when engine is running. Fuel pump relay is the recommended source.
Black	Electronics Ground	Ground	Ground where interfacing device is grounded
White	Heater Ground	Ground	Ground to chassis or engine block
Green	High Performance Analog Output	Interfacing device; ECU/Gauge/datalogger/etc…	Default output is linear output; 0[v] @ 0.68 [Lambda] Linear to 5[v] @ 1.36 [Lambda], equivalent to 10-20 [AFR] for gasoline fuel
Brown	Standard Performance Analog Output		Default output is heater status; 0[v] = waiting for trigger before heating sensor, 1[v] = sensor heatup started, 2[v] = sensor heatup complete and sensor heater is in closed loop control
Blue	CAN High		
Purple	CAN Low		
Orange	UART TX	"Rx-Orange" on USB converter	
Yellow	UART RX	"Tx-Yellow" on USB converter	
Grey	UART Ground	"Gnd-Grey" on USB converter	

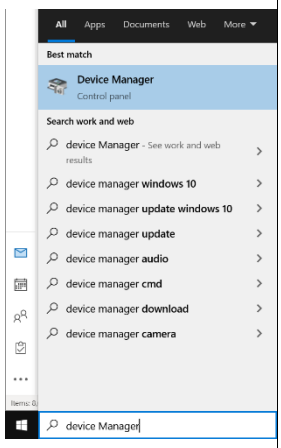
6. USB Communications

Spartan 3 comes with a USB converter to provide USB communications with your computer. The converter is based on the popular FTDI chipset so most operating systems already have the driver preinstalled.

7. Windows 10 serial terminal

Electronics Ground (Black), Heater Ground (white), and Power (Red) must be connected and Spartan 3 powered to enter serial commands. Spartan 3 cannot be powered through USB.

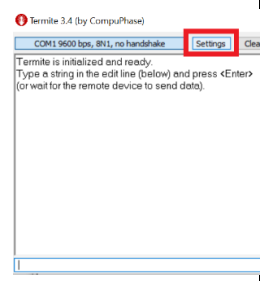
The recommended serial terminal is Termite, https://www.compuphase.com/software_termite.htm, please download and install the complete setup.



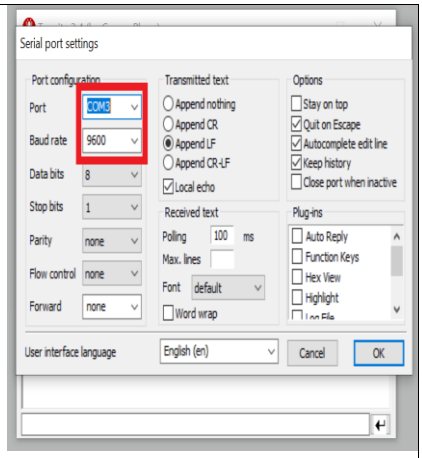
In windows 10 search bar, please type “Device Manager” and open it.



Spartan 3 USB converter will show up as “USB Serial Port”, in this example “COM3” is assigned to Spartan 3.



In Termite, click “Settings”



Make sure the Port is correct and that the Baud rate is “9600”.

8. Serial Commands

Electronics Ground (Black), Heater Ground (white), and Power (Red) must be connected and Spartan 3 powered to enter serial commands. Spartan 3 cannot be powered through USB.

Serial Command	Usage Note	Purpose	Example	Factory Default
GETHW		Gets Hardware Version		
GETFW		Gets Firmware version		
SETTYPEx	If x is 0 then Bosch LSU 4.9 If x is 1 then Bosch LSU ADV	Sets LSU sensor type	SETTYPE1	x=0, LSU 4.9
GETTYPE		Gets LSU sensor type		
SETCANFORMATx	x is an integer 1 to 3 character long. x=0; default x=1; Link ECU x=2; Adaptronic ECU x=3; Haltech ECU x=4; % Oxygen*100 x=5; Extended CAN format		SETCANFORMAT0	x=0
GETCANFORMAT		Gets CAN format		
SETCANIDx	x is an integer 1 to 4 characters long	Sets 11 bit CAN id	SETCANID1024 SETCANID128	x=1024
GETCANID		Gets 11 bit CAN id		
SETCANBAUDx	x is an integer 1 to 7 characters long	Sets CAN Baud Rate	SETCANBAUD1000000 will set CAN Baud rate to 1Mbit/s	X=500000, 500kbit/s
GETCANBAUD		Gets CAN Baud Rate		
SETCANRx	If x is 1 the resistor is enabled. If x is 0 the resistor is disabled	Enable/Disable CAN Termination Resistor	SETCANR1 SETCANR0	x=1, CAN term Res Enabled
GETCANR		Gets CAN Term Res State; 1=enabled, 0=disabled		

SETAFRMxx.x	xx.x is a decimal exactly 4 characters long including decimal point	Sets AFR Multiplier for Torque app	SETAFM14.7 SETAFM1.00	xx.x=14.7
GETAFRM		Gets AFR Multiplier for Torque app		
SETLAMFIVEVx.xx	x.xx is a decimal exactly 4 characters long including decimal point. Minimum value is 0.60, maximum value is 3.40. This value can be higher or lower than the SETLAMZEROV value.	Sets Lambda at 5[v] for the linear output	SETLAMFIVEV1.36	x.xx=1.36
GETLAMFIVEV		Gets the Lambda at 5[v]		
SETLAMZEROVx.xx	x.xx is a decimal exactly 4 characters long including decimal point. Minimum value is 0.60, maximum value is 3.40. This value can be higher or lower than the SETLAMFIVEV value.	Sets Lambda at 0[v] for the linear output	SETLAMZEROV0.68	x.xx=0.68
GETLAMZEROV		Gets Lambda at 0[v]		
SETPERFx	If x is 0 then standard performance of 20ms. If x is 1 then high performance of 10ms. If x is 2 then optimize for lean operation.		SETPERF1	x=1, high performance
GETPERFx		Gets performance		
SETSLOWHEATx	If x is 0 then sensor is heated at normal rate during initial power up. If x is 1 then sensor is heated at 1/3 the normal rate during initial power up. If x is 2 then wait, a max of 10 min, for MS3 CAN RPM signal before heating. If x is 3 then wait, a max of 10 min, for exhaust gas to heat sensor to 350C before heating. Requires Firmware 1.05 and above		SETSLOWHEAT1	x=0, normal sensor heatup rate
GETSLOWHEAT		Gets slowheat setting		
MEMRESET		Reset to factory settings.		
SETLINOUTx.xxx	Where x.xxx is a decimal exactly 5 characters long including decimal point, greater than 0.000 and less than 5.000. Linear Output will resume normal operation on reboot.	Allows the user to set the High Perf Linear Output to a specific voltage	SETLINOUT2.500	
DOCAL	Free Air Calibration is no longer supported. It causes more problems than it solves. Spartan 3 is designed to be accurate without free air calibration.	Do Free Air Calibration and display the value. A perfect sensor will have a value of 1.00.		
GETCAL	No longer supported	Gets Free Air Calibration value		
RESETCAL	No longer supported	Resets Free Air Calibration value to 1.00 (factory)		
SETCANDRx	x is an integer 1 to 4 characters long Requires Firmware 1.04 and above	Sets CAN Data Rate in hz	For optimal performance, CAN Data Rate should be 200hz	x=50
GETCANDR	Requires Firmware 1.04 and above	Gets CAN Data Rate		
SETNBMODEx	If x is 0 then Standard Performance Analog Output is Simulated Narrowband. If x is 2 then Standard Performance Analog Output is Heater Status	Selects the function of the Standard Performance Analog Output between Simulated Narrowband and Heater Status	SETNBMODE0	x=2, Heater Status
GETNBMODE		Gets Standard Performance Analog Output mode		
SETNBSWLAMx.xxx	x.xxx is a decimal exactly 5 characters long including decimal point. Requires Firmware 1.08 and above	Sets the Simulated Narrowband Switch Point in Lambda	SETNBSWLAM1.005	x.xxx=1.000
GETNBSWLAM		Gets the Simulated Narrowband switch point in Lambda		

*All commands are in ASCII, case does not matter, spaces do not matter.

9. CAN Bus Protocol Default Format (Lambda)

For %O2 CAN Format please see “Spartan 3 and Spartan 3 Lite for Lean Burn and Oxygen Metering Applications.pdf”

Spartan 3’s CAN Bus operates with 11 bit addressing.

Default CAN Baud rate is 500kbit/s

Default CAN Termination resistor is enabled, this can be changed by sending “SETCANRx” serial command.

Default CAN Id is 1024, this can be changed by sending “SETCANIDx” serial command.

Data Length (DLC) is 4.

Default Data Rate is 50 hz, data is sent every 20[ms], this can be changed by sending “SETCANDRx” serial command.

Big-endian for all data

Data[0] = Lambda x1000 High Byte

Data[1] = Lambda x1000 Low Byte
Data[2] = LSU_Temp/10
Data[3] = Status
Lambda = (Data[0]<<8 + Data[1])/1000
Sensor Temperature [C] = Data[2]*10

Status

Status Value (decimal)	Meaning
0	Reserved
1	Waiting for trigger before heating up
2	Sensor is heating up
3	Sensor in normal operation
4+	Reserved

10. Supported CAN Formats

CAN Format Name	CAN Format Serial Command	CAN Id Serial Command	CAN BAUD Rate Serial Command	Note
Link ECU CAN Format	SETCANFORMAT1	SETCANID950	SETCANBAUD1000000	Read "Link G4+ ECU CAN Bus Configuration for Spartan 3.pdf"
Adaptronic ECU CAN Format	SETCANFORMAT2	SETCANID1024 (Default from factory)	SETCANBAUD1000000	
MegaSquirt 3 ECU CAN Format	SETCANFORMAT0 (Default from factory)	SETCANID1024 (Default from factory)	SETCANBAUD500000 (Default from factory)	Read "Interfacing Spartan 3 to MegaSquirt 3.pdf"
Haltech ECU CAN Format	SETCANFORMAT3	Not required	SETCANBAUD1000000	Spartan 3 Emulates Haltech WBC1 wideband controller
YourDyno Dyno Controller CAN Format	SETCANFORMAT0 (Default from factory)	SETCANID1024 (Default from factory)	SETCANBAUD1000000	
MaxxECU	SETCANFORMAT0 (Default from factory)	SETCANID1024 (Default from factory)	SETCANBAUD500000 (Default from factory)	Read "Interfacing Spartan 3 to MaxxECU.pdf"
Extended CAN Format	SETCANFORMAT5			Read "Spartan 3 Extend CAN Format.pdf" Requires Firmware 1.08 and above.

11. CAN Termination Resistor:

Suppose we call the ECU; Master, and devices that send/receive data to/from the ECU we call; Slave (Spartan 3, digital dashboard, EGT controller, etc…). In most applications there is one Master (ECU) and one or more Slaves that all share the same CAN Bus. If Spartan 3 is the only Slave on the CAN Bus then the CAN Termination Resistor on Spartan 3 should-be enabled using the serial command "SETCANR1". By default the CAN Termination Resistor on Spartan 3 is enabled. If There are multiple Slaves, the Slave that is farthest from the Master (based on wire length) should have the CAN Termination Resistor enabled, all other Slaves should have their CAN Termination Resistor disabled/disconnected. In practice; it often does not matter if the CAN Termination Resistors are properly set, but for highest reliability the CAN Termination Resistors should be properly set.

12. Bootloader

When Spartan 3 is powered up without the LSU Heater Ground, white wire, connected it will enter bootloader mode. Powering up Spartan 3 with the Heater Ground connected will not trigger the bootloader and Spartan 3 will work as normal.

Warranty

14Point7 warrants Spartan 3 to be free from defects for 2 years.

Disclaimer

14Point7 is liable for damages only up to the purchase price of its products. 14Point7 products should not be used on public roads.